

Department of Nano Science Technology

Faculty of Technology

Wayamba University of Sri Lanka

Centralized Workshops and Laboratory Allocation Platform

USE-CASE DESCRIPTION

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1. Introduction

This section describes the main use cases of SYNCLAB, a desktop-based system developed to manage laboratory and workshop allocation requests within the Faculty of Technology. The system provides a student interface for submitting and managing requests and an administrator interface for reviewing and managing allocation records. It aims to reduce paperwork and scheduling difficulties associated with the manual allocation process.

2. Actors and Their Responsibilities

Actor	Description	Responsibilities
Student / User	A student who needs a laboratory or workshop for an academic activity or project.	Select a facility, date and time slot; enter required details; submit a request; keep the Allocation Key; use the key to retrieve, update or delete the request.
Administrator (Technical Officer / Laboratory In-Charge)	The authorized person responsible for managing requests for a particular facility.	Access the facility dashboard using its administrator passcode; view requests and statuses; manage records; add remarks; prepare email drafts; print allocation information.
HOD and Dean	External approval authorities involved in the faculty approval process.	Receive approval-related communications from responsible staff and provide the required approval or rejection according to faculty procedures. They do not directly operate the SYNCLAB interface in the described workflow.

Note: A separate Guest actor is not included because the described SYNCLAB workflow does not provide a guest-specific function. Student functions are accessed through the Student / User interface, while administrative functions are protected by passcodes.

3. Main Use Cases

UC-01: Submit a New Allocation Request

Actor(s): Student / User

Purpose: To request a date and time slot for using a selected laboratory or workshop.

Precondition: The application is open and the student can access the student interface.

Main Flow:

- The student selects “New Allocation” from the welcome screen.
- The student selects the required laboratory or workshop.
- The student chooses an available date and one of the displayed time slots.
- The system displays the selected facility, date and time for confirmation.
- The student enters the required details: name, index number, department, email address, supervisor name and project name. A description is optional.
- The student confirms the request.
- The system checks whether the selected facility, date and time slot already has an active allocation.
- If the slot is available, the system saves the request with Pending status and displays a unique Allocation Key.

Postcondition: The request is stored as Pending, and the student receives an Allocation Key to retrieve it later.

Alternative / Exception: If the slot is already reserved, the system does not create a duplicate active allocation. The student must select another available slot.

UC-02: Retrieve and Edit an Existing Request

Actor(s): Student / User

Purpose: To access a previously submitted request and update its information.

Precondition: The student has a valid Allocation Key for an existing request.

Main Flow:

- The student selects “Edit Allocation” from the welcome screen.

- The student enters the Allocation Key.
- The system searches for the matching allocation record.
- If a matching record is found, the system displays the existing facility, date, time and submitted details.
- The student changes the required information.
- The student confirms the changes.
- The system checks the availability of the newly selected facility, date and time slot.
- The system saves the updated request and sets its status to Pending for review.

Postcondition: The allocation record contains the updated information and has Pending status.

Alternative / Exception: If the key is invalid, the request cannot be retrieved. If the newly selected slot is unavailable, the update cannot be completed using that slot.

UC-03: Review and Manage Allocation Requests

Actor(s): Administrator (Technical Officer / Laboratory In-Charge)

Purpose: To view and manage requests submitted for a specific laboratory or workshop.

Precondition: The administrator has the correct passcode for the selected facility.

Main Flow:

- The administrator selects the relevant facility and opens its administration dashboard.
- The system requests the facility administrator passcode.
- After successful verification, the system displays the facility dashboard, calendar and allocation table.
- The administrator reviews the Allocation Key, student information, requested date and time, description and current status.
- The administrator may select a record to edit or delete it, add or update a remark, view allocation dates, or print the allocation list.
- The administrator reviews the request and updates its status according to the system workflow.

Postcondition: The administrator can view and manage allocation records for the selected facility. Saved changes are reflected in the system.

Alternative / Exception: If the passcode is incorrect, access to the facility administration dashboard is not granted.

UC-04: Prepare Approval or Rejection Email Draft

Actor(s): Administrator; HOD and Dean are supporting external actors

Purpose: To prepare an email related to the approval or rejection of an allocation request.

Precondition: The administrator has accessed the relevant facility dashboard and selected an allocation record.

Main Flow:

- The administrator selects an allocation record.
- For an approval-related message, the administrator selects “Pending Email Draft”. For a rejection-related message, the administrator selects “Rejection Email Draft”.
- The system prepares an email draft using the available allocation details and configured email addresses.
- The draft opens in the computer's default email application.
- The administrator checks the message and recipients, makes any required changes, and sends the email manually.

Postcondition: An email draft is prepared. The email is sent only if the administrator manually sends it through the external email application.

Alternative / Exception: If the email application or email settings are unavailable or not configured, the administrator may need to configure them before sending.

UC-05: Configure System Settings

Actor(s): Authorized Administrator

Purpose: To manage application settings and facility administrator passcodes.

Precondition: The authorized administrator has the master passcode required to access Settings.

Main Flow:

- The administrator selects “Settings” from the welcome screen.
- The system requests the master passcode.
- After successful verification, the system displays the available settings.
- The administrator may change the application theme, select a background wallpaper, or change an individual facility administrator passcode.
- The administrator saves the selected changes.
- The system applies or stores the updated settings.

Postcondition: The selected settings are updated and available to the application.

Alternative / Exception: If the master passcode is incorrect, access to the settings is denied.

4. Use-Case Diagram

The following simplified overview shows the relationship between the actors and the main use cases. It can be used as a guide when preparing a graphical UML use-case diagram.

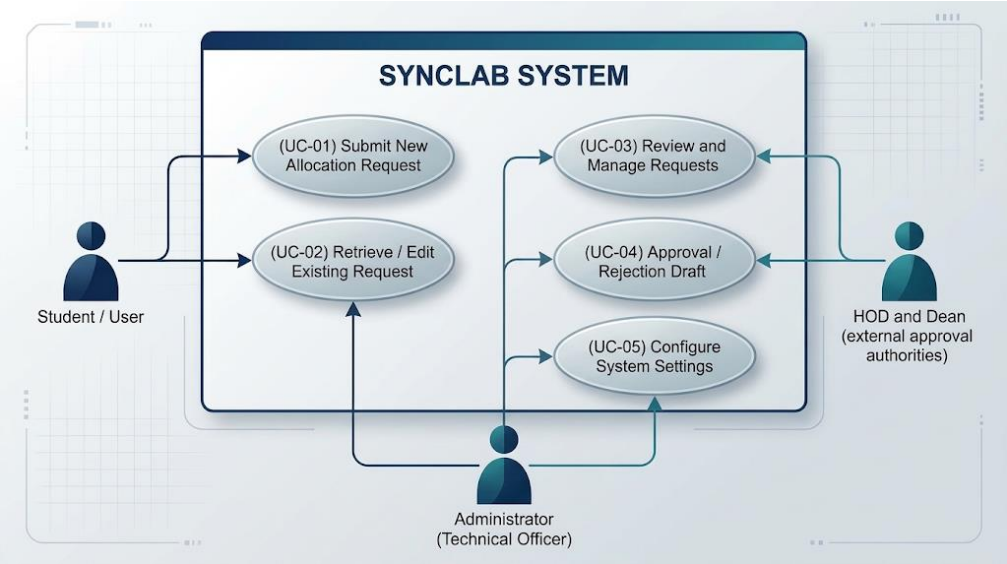


Figure 1 Use-Case Diagram

5. Summary

The use cases describe the main activities supported by SYNCLAB, from submitting a student request to administrative review and approval-related communication. Students can submit and retrieve requests using an Allocation Key, while authorized administrators can manage facility records and prepare email drafts. System settings are protected by a master passcode. The HOD and Dean remain external approval authorities, and email drafts must be reviewed and sent manually by staff.